Riley Beenders

Lead, R&D Engineer

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RileyBeenders - MyFolio Project

PROFILE

Lead R&D Engineer with prior CTO experience developing and commercializing complex electromechanical systems and additive manufacturing equipment. Skilled in CATIA, SolidWorks, fixture and tooling design, DFM, production workflow optimization, and cross-functional mechanical, electrical, firmware, and software integration. Experienced taking products from prototype through production while improving manufacturability, reliability, and serviceability.

EXPERIENCE

Proteor / Proteor Printing Solutions

Lead, Research & Development Engineer — *Coplay, PA* · **Jan 2025 - Present**

Proteor acquired Filament Innovations in January, 2025 and my role evolved from CTO to Lead R&D Engineer. I continue to carry out my former CTO responsibilities, now with expanded oversight of the East Coast engineering team and involvement across Proteor's global product portfolio.

- Led process development, design, and implementation for the ICARUS-Lite, a compact 3D printer for digital prosthetic and orthotic production. Delivered custom-designed AC heaters with dual-voltage 120/240 VAC capabilities, increasing internal production by 30%.
- Led the ICARUS-Lite from product development through commercialization, applying design for assembly principles to support manufacturability and contributing to increased revenue and the addition of approximately five new roles.
- Developed and validated an automated LabVIEW system that collects nine data points every 15 ms from multiple laser sensors over Modbus and charts tolerance data for internal quality control and end-customer use.
- Led the implementation of Microsoft Business Central across the division's product portfolio, integrating data with global Proteor locations to streamline sales and inventory management.
- Led the implementation of ProteorPrint OS, a network-accessible remote operation and diagnostics interface with live video, AI-assisted alerts, and material-waste metrics, reducing equipment downtime and enhancing remote support.
- Performed and documented risk analysis covering safety, operational, and software risks for ICARUS and ICARUS-Lite products along with the ProteorPrint cloud platform, implementing mitigations applicable to EU and U.S. standards.
- Collaborated with another additive manufacturer to develop and validate a reliable, cost-effective hotend that met material-specific requirements.
- Worked alongside production, sales, and customer-service teams to support the product line, providing hardware, electronics, motion-control, software troubleshooting, and after-sales technical support, which increased customer-satisfaction.
- Converted the Filament Innovations POSEIDON and ICARUS platforms from servo-driven ballscrews to a standardized linear-motor architecture, achieving 1-micron precision at 1 m/s while moving a 30 kg payload. The system reached a maximum speed of 4 m/s.
- Continued development of our internal IT infrastructure and standardized software and internal processes within our division of Proteor

Filament Innovations

Chief Technology Officer — *Whitehall, PA* · **Jan 2023 - Jan 2025**

Acquired by Proteor and is recognized globally as 'Proteor Print'.

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while retaining hands-on responsibility for mechanical, electrical, firmware, and software systems. Used CAD software, including CATIA, to support engineering design and development.
- Led the design and launch of the Gen3 POSEIDON, combining a 1 m^3 build volume, filament-and-pellet dual extrusion, braked NEMA 34 Z drives, industrial servo XY motion, and a custom gantry while supporting precision assembly and production requirements.
- Designed and hosted a redundant Kubernetes cluster for a self-developed ERP/CRM that managed orders, inventory, tasks, projects, and customer interactions, with dynamic BOM generation linked to purchasing, which cut order-processing time by almost 65% and increased inventory accuracy.
- Researched and developed high-speed linear motor systems that moved a 30 kg payload at 3 m/s with 3-micron accuracy; validated test results against performance targets. Worked with oversea suppliers to coordinate manufacturing and supported deployment in upcoming ProteorPrint models.

Lead Electro-Mechanical Engineer — *Whitehall, PA* · **Jun 2022 - Jan 2023**

- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling with local fabricators for large-radius aluminum-panel bends.

- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging while supporting larger, modular machines.
- Developed independently testable electronics enclosures that could be installed in minutes and coordinated custom harnesses, panel designs, and hardware improvements with vendors.

Industrial Engineer *Whitehall, PA* · **Apr 2021 - Jun 2022**

- Designed and released KRATOS, a machine that prints high-strength above and below-knee prosthetic sockets faster.
- Developed next-generation ICARUS and POSEIDON models with higher-performance components and expanded motion safety through NEMA 34 motors, planetary gearboxes, and power-failure brakes.
- Designed THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry printer for a client producing the world's largest 3D-printed statue, and managed CAD, BOM, sourcing, fabrication, assembly, firmware tuning, support, and training.

Engineering Technician *Whitehall, PA* · **May 2020 - Apr 2021**

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized print parameters while documenting design limitations and opportunities for product improvement.

RJB.Engineering

Owner & Engineering Consultant *Pennsylvania* · **2015 - Present**

- Deliver project-based engineering support for automotive clients utilizing CATIA and MATLAB.
- Produced 3D-printed plastic face shields using a custom designed printer, supplying local healthcare providers during the COVID-19 pandemic to meet urgent protective-equipment needs.

Blue Mountain Ski Resort

Ski Lift Operator *Palmerton, PA* · **Nov 2019 - Jan 2021**

- Operated lifts and assisted thousands of guests with ticketing, weather, and slope information.
- Performed motor and control safety checks and supported the maintenance team during lift breakdowns.

Northampton Area School District

Computer / Network Technician *Northampton, PA* · **Jun 2016 - Aug 2019**

- Assisted with my school district's network infrastructure, installing over 750 wireless access points across six school buildings, which increased Wi-Fi coverage and reduced connectivity issues.
- Accelerated deployment of more than 2,500 Chromebooks by developing an Arduino-based tool to automate setup-data entry.
- Managed the final distribution of devices with school administrators, creating detailed documentation and checklists that ensured all students received the correct equipment.

EDUCATION

Penn State University **Dec 2024**

BS, Electro-Mechanical Engineering

CERTIFICATIONS

- MATLAB OnRamp: MathWorks | Issued July 2026
- Simulink OnRamp: MathWorks | Issued July 2026
- Simscape OnRamp: MathWorks | Issued July 2026
- Machine Learning Onramp: MathWorks | Issued July 2026
- Lean Six Sigma Yellow Belt: Educate 360 | Issued July 2025
- Lean Six Sigma White Belt: Educate 360 | Issued June 2025
- Certificate in Engineering Design and Tools: Penn State University | Issued Spring 2023
- Mechanical Engineering Certificate: Lafayette College | Issued Spring 2019

SKILLS

- **Engineering**: Design for Manufacturing, Design For Assembly (DFA), Electrical Layout and Schematic Design, Engineering Documentation and BOM Control, Geometric Dimensioning & Tolerancing (GD&T), Lean Manufacturing, Product Development and Commercialization, Product Safety and Regulatory Compliance, Process Development, Production Workflow Optimization, R&D Prototyping & Validation, Technical Writing and Documentation, Vendor Communication and Cost Negotiation, Worldwide Project Collaboration, Work Instructions And SOP Writing
- **Design & Software**: Additive Manufacturing, Autodesk Inventor, CATIA, Fixture and Tooling Design, Fusion 360, GitHub, LabVIEW, LM Studio, Macros, MATLAB, oMLX, Python, Rapid Prototyping, Siemens NX, SolidWorks, Visual Studio Code
- **Controls & Automation**: Allen Bradley PLCs, Automation Equipment Development, Data Logging and Process Analysis, Git Version Control, Workflow Automation
- **Infrastructure & Systems**: Domain-Based Application Access, DNS and Subdomain Configuration, Docker and Containerized Applications, Proxmox Virtualization, Reverse Proxy and Firewall Routing, Self-Hosted Infrastructure, Server Network and VLAN Configuration, Virtual Machine and Container Management, Windows/Linux Systems Administration